

Quantum Mechanics from Graph Structure

Peter Groom
Dawn Field Institute

May 2026

On deriving the Hilbert space, Born rule, Schrödinger equation, Bell violation, and decoherence from ADE automorphisms without quantum postulates

Peter Groom, Dawn Field Institute

PACSeries Paper 11

Date: May 2026

Version: 1.0 (Draft)

Abstract

We derive the mathematical structure of quantum mechanics — Hilbert spaces, the Born rule, the Schrödinger equation, Bell inequality violation, and decoherence — from the automorphism groups of ADE Dynkin diagrams, without postulating any quantum axioms.

The central construction is the orbit Hilbert space $\mathcal{H} = L^2(V/\text{Aut}(G))$: the space of square-integrable functions on vertex orbits under graph automorphisms. For a graph G with vertices V and automorphism group $\text{Aut}(G)$, the orbit basis vectors $|O_i\rangle$ span a Hilbert space whose dimension equals the number of distinct orbits. This construction resolves the positive semi-definite (PSD) degeneracy that limited Milestone 13's complement framework: the orbit Gram matrix is the identity for all ADE types, by construction.

The results: (1) The Born rule $P(O_i) = |O_i|/n$ follows from orbit measure: the probability of finding the system in orbit O_i is the fraction of vertices in that orbit. (2) D_4 (the graph with Dynkin type D_4 , automorphism group S_3) is the only ADE type with non-abelian automorphisms among types with rank ≤ 8 . Non-abelian $\Rightarrow$ non-commuting orbit observables $\Rightarrow$ Robertson uncertainty bound > 0 . Quantum uncertainty has a unique graph-theoretic origin. The non-commutativity measure is $\mathcal{NC} = 1.2247$ for D_4 and exactly 0 for all other ADE types tested. (3) Measurement is gauge fixing: projecting onto an orbit eigenstate selects a representative from a gauge equivalence class, and the real projectors necessarily lose phase information, the SEC arrow applied to measurement. (4) Under the orbit-Laplacian (graph-derived) SEC rotation, the CHSH Bell parameter rises from $S = 2.000$ (fixed basis) to $S = 2.514$ on $D_4 \times D_4$ product graphs — a clear violation at 88.8% of the Tsirelson bound. The orbit space saturates the Tsirelson bound ($S = 2\sqrt{2} = 2.828$ to 4×10^{-16}) only under the ideal $\text{SU}(2)$ generator $\sigma_y/2$, confirming that it supports the full quantum correlation range even though the graph-derived generator alone does not reach saturation. Graphs with nontrivial (abelian) automorphisms such as A_n can exceed the classical bound; graphs with trivial automorphisms (E_7 , E_8) do

not violate Bell inequalities at all. (5) The Schrödinger equation emerges: the orbit Laplacian $H = B^T L B$ (where L is the graph Laplacian and B is the orbit basis) serves simultaneously as the Hamiltonian, the Bell measurement generator, and the path-counting matrix. Time evolution $U(t) = e^{-iHt}$ is unitary, automorphism-equivariant ($[P_g, H] = 0$), and energy-conserving — verified to machine precision (2×10^{-15}). (6) The Feynman path integral is a sum over graph hops: the propagator $K(t) = e^{-iHt} = \sum_n (-iH)^n t^n / n!$ where H^n counts n -hop paths on the orbit graph. Multi-path interference on 3-orbit systems (A_5) shows visibility 0.87. (7) The quantum Zeno effect freezes orbit transitions: $P_{\text{survive}}(N = 1000) = 0.997$ on D_4 , with anti-Zeno enhancement at $N = 2$ (26% above free decay) — the crossover rate encodes the spectral gap, hence the topology. (8) Decoherence from orbit-environment coupling follows an exact formula $|\rho_{12}(t)| = \frac{1}{2} |\langle 0_{\text{env}} | e^{-2i\lambda H_{\text{env}} t} | 0_{\text{env}} \rangle|$, verified to 1.7×10^{-15} . The Fermi golden rule $\Gamma^2 = \lambda^2 \cdot \text{Var}(H_{\text{env}})$ holds across 6 ADE topologies with 0.04% spread. The orbit basis is the pointer basis: orbit eigenstates maintain purity 1.000000 under dephasing while superpositions drop to 0.51.

We report 62/66 pass (94%) across M14 (40/44) and P13–P16 (22/22). The 4 failures: orbit-level interference is algebraic not positional (superpositions of disjoint orbits cannot produce vertex-level cross-terms), and the graph double-slit experiment achieves only 1/4 tests because interference requires same-orbit overlap, not cross-orbit superposition.

Zero contradictions with Milestones 1–13. 12 predictions from M14, plus 4 new predictions from P13–P16. The framework makes quantum mechanics a consequence of graph symmetry rather than a separate axiom system.

Keywords: quantum mechanics, ADE graphs, orbit Hilbert space, Born rule, Bell inequality, Schrödinger equation, path integral, Zeno effect, decoherence, einselection, automorphism group, Dawn Field Theory

Epistemic status. This paper is part of the PACSeries, an evolving research program rather than a finished theory (see the series note *How to Read the PACSeries*). Its quantitative claims are tiered by derivation type — **A** structural, **B** identified, **C** pattern-matched — and labeled in the text. Failures are reported rather than omitted, and framework-level corrections are logged in the Epistemic Corrections Registry in the repository. Every headline result is reproducible from the `Code/` and `Data/` in this package (provenance in `trace.yaml`).

1. The quantum axiom problem

Quantum mechanics rests on axioms that are assumed, not derived:

1. States live in a Hilbert space $\mathcal{H}$
2. Observables are Hermitian operators on $\mathcal{H}$
3. Measurement probabilities follow the Born rule $P = |\langle \psi | \phi \rangle|^2$
4. Time evolution is unitary: $|\psi(t)\rangle = e^{-iHt} |\psi(0)\rangle$
5. Composite systems use the tensor product: $\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B$

These axioms work well. But they are postulated, not explained. Why Hilbert space and not some other mathematical structure? Why the Born rule and not some other probability assignment? Why unitary evolution?

This paper derives all five from graph theory. The ADE Dynkin diagrams — the same classification that produced gauge groups (Paper 4), spacetime (Paper 10), and the force hierarchy

(Paper 7) — also produce quantum mechanics. The Hilbert space is the orbit space; the Born rule is counting; unitary evolution is the graph Laplacian exponential; the tensor product is the Cartesian product of graphs.

2. The orbit Hilbert space

2.1 Construction

Let G be an ADE Dynkin diagram with vertex set $V = \{v_1, \dots, v_n\}$ and automorphism group $\text{Aut}(G)$.

The automorphism group partitions V into orbits $\{O_1, O_2, \dots, O_d\}$, where vertices in the same orbit are related by graph symmetries. Two vertices v_i, v_j are in the same orbit iff there exists $g \in \text{Aut}(G)$ with $g(v_i) = v_j$.

Define the orbit basis:

$$|O_k\rangle = \frac{1}{\sqrt{|O_k|}} \sum_{v \in O_k} |v\rangle$$

These are orthonormal: $\langle O_i | O_j \rangle = \delta_{ij}$.

The orbit Hilbert space is $\mathcal{H}_{\text{orb}} = \text{span}\{|O_1\rangle, \dots, |O_d\rangle\}$, with dimension $d = \text{number of orbits}$.

2.2 Examples

ADE type	n vertices	$ \text{Aut}(G) $	Aut type	d orbits	Orbit structure
A_2	2	2	$\mathbb{Z}_2$	1	All equivalent
A_3	3	2	$\mathbb{Z}_2$	2	{ends}, {center}
A_4	4	2	$\mathbb{Z}_2$	2	{outer pair}, {inner pair}
A_5	5	2	$\mathbb{Z}_2$	3	{ends}, {inner}, {center}
D_4	4	6	S_3	2	{hub}, {3 leaves}
E_6	6	2	$\mathbb{Z}_2$	4	By mirror symmetry
E_7	7	1	trivial	7	Every vertex distinct
E_8	8	1	trivial	8	Every vertex distinct

D_4 has 2 orbits (hub + leaves) with automorphism group S_3 (permutations of 3 leaves). This makes the orbit Hilbert space 2-dimensional, a two-level (qubit) system.

2.3 Resolution of PSD degeneracy

M13 (Paper 10) found that the complement Gram matrix is positive semi-definite but not positive definite for some ADE types — a fundamental obstruction to defining a metric on complement space.

The orbit construction resolves this completely. The orbit Gram matrix $G_{ij} = \langle O_i | O_j \rangle = \delta_{ij}$ is the identity matrix for all ADE types, by orthonormality. PSD degeneracy is not a defect but a signal that the correct inner product lives on orbits, not vertices.

3. The Born rule from orbit measure

3.1 Derivation

For a system in state $|\psi\rangle = \sum_k c_k |O_k\rangle$ (normalized: $\sum_k |c_k|^2 = 1$), the probability of finding the system in orbit O_k is:

$$P(O_k) = |c_k|^2$$

For the uniform state $|\psi_{\text{uniform}}\rangle = \frac{1}{\sqrt{n}} \sum_v |v\rangle$, this gives:

$$P(O_k) = \frac{|O_k|}{n}$$

This is the Born rule. The probability of an outcome equals the orbit volume, the fraction of graph vertices in that orbit. It is not postulated but follows from counting gauge-equivalent configurations.

3.2 Verification

Tested on all ADE types with rank ≤ 8 (M14 exp_03). For every type, Born probabilities from orbit measure match direct computation to machine precision. The result is algebraic, not approximate.

For D_4 : $P(\text{hub}) = 1/4$, $P(\text{leaves}) = 3/4$. This is the "quantum" probability of finding the system at the hub vs a leaf — derived from the graph structure, not from a measurement postulate.

4. Measurement as gauge fixing

4.1 The mechanism

Measurement in quantum mechanics collapses a superposition to an eigenstate. In the orbit framework, measurement is gauge fixing: selecting a specific representative from a gauge equivalence class (M14 exp_04).

The orbit projector $P_{O_k} = |O_k\rangle\langle O_k|$ is a real operator. Projecting a complex state $|\psi\rangle$ onto P_{O_k} necessarily discards phase information:

$$P_{O_k}|\psi\rangle = c_k|O_k\rangle$$

The lost phase is the gauge degree of freedom — which specific vertex within the orbit is selected. This information is physically meaningless (all vertices in an orbit are equivalent), so "measurement" is the removal of gauge redundancy.

4.2 The SEC arrow

The projector P_{O_k} is real. The state $|\psi\rangle$ may be complex (via SEC complexification). Real projection of a complex state is irreversible — you cannot recover the phase from the projected result.

This irreversibility is the SEC arrow applied to measurement. The second law of thermodynamics and the measurement postulate have the same graph-theoretic origin.

5. Quantum uncertainty from D_4 triality

5.1 The uniqueness result

Among all ADE types with rank ≤ 8 , only D_4 has a non-abelian automorphism group (S_3 , the symmetric group on 3 elements). Non-abelian means non-commuting group elements, which means non-commuting orbit observables.

Define the non-commutativity measure:

$$\mathcal{NC}(G) = \sqrt{\frac{1}{|\text{Aut}|^2} \sum_{g,h \in \text{Aut}} \|[g, h]\|_F^2}$$

Results (M14 exp_07):

ADE type	Aut type	$\mathcal{NC}$
A_n ($n \leq 8$)	$\mathbb{Z}_2$ or trivial	0
D_4	S_3	1.2247
D_n ($n > 4$)	$\mathbb{Z}_2$	0
E_6	$\mathbb{Z}_2$	0
E_7, E_8	trivial	0

D_4 is uniquely non-commutative. Quantum uncertainty — the Robertson bound $\Delta A \cdot \Delta B \geq \frac{1}{2} |\langle [A, B] \rangle|$ — requires $[A, B] \neq 0$, which requires non-abelian automorphisms, which requires D_4 (M14 exp_08).

5.2 Physical interpretation

D_4 has three equivalent leaves connected to a hub. The S_3 automorphism permutes these leaves. Two different permutations (e.g., swap leaves 1,2 vs swap leaves 1,3) do not commute — the order matters. This non-commutativity propagates to orbit observables, creating genuine quantum uncertainty.

The number 3 is special: $S_2 = \mathbb{Z}_2$ (abelian), but S_3 is the smallest non-abelian symmetric group. D_4 triality — the three-fold symmetry of the Dynkin diagram — is the unique source of quantum non-commutativity in ADE.

6. Entanglement from product graphs

6.1 Construction

For two systems with graphs G_A and G_B , the composite system lives on the Cartesian product graph $G_A \square G_B$. The vertex set is $V_A \times V_B$, with edges between (a_1, b) and (a_2, b) iff $a_1 \sim a_2$ in G_A , and between (a, b_1) and (a, b_2) iff $b_1 \sim b_2$ in G_B .

The orbit Hilbert space of the product is the tensor product of orbit spaces:

$$\mathcal{H}_{\text{orb}}(G_A \square G_B) = \mathcal{H}_{\text{orb}}(G_A) \otimes \mathcal{H}_{\text{orb}}(G_B)$$

The tensor product structure is a theorem about Cartesian products of graphs, not a postulate (M14 exp_09).

6.2 Bell states

On $D_4 \times D_4$ (2 orbits each, so $2 \times 2 = 4$ -dimensional orbit space), the maximally entangled Bell state is:

$$|\Psi\rangle = \frac{1}{\sqrt{2}}(|O_1\rangle_A |O_1\rangle_B + |O_2\rangle_A |O_2\rangle_B)$$

The entanglement entropy is $S = \ln 2 = 0.6931$ (maximal for a 2-qubit system). The purity of the reduced state is $\text{Tr}(\rho_A^2) = 0.5$ (maximally mixed).

7. Bell inequality violation from topology

7.1 The CHSH test on orbit space

The CHSH Bell parameter is $S = E(a, b) - E(a, b') + E(a', b) + E(a', b')$, where $E(a, b) = \langle \Psi | A(a) \otimes B(b) | \Psi \rangle$ are correlations for measurement settings a, b .

Classical hidden variable theories give $|S| \leq 2$. Quantum mechanics allows $|S| \leq 2\sqrt{2} \approx 2.828$ (Tsirelson bound).

On $D_4 \times D_4$ orbit space (P13):

- **Fixed basis** (no rotation): $S = 2.000$ — classical bound, no violation
- **Orbit-Laplacian (SEC-rotated) basis**: $S = 2.514$ — a clear violation, 88.8% of the Tsirelson bound
- **Ideal SU(2) generator** ($\sigma_y/2$, not graph-derived): $S = 2\sqrt{2} = 2.828$ — Tsirelson bound, saturated to 4×10^{-16}

The graph-derived rotation is generated by $R(\theta) = e^{i\theta G}$ with G from the orbit Laplacian — SEC complexification applied to measurement. It produces a genuine Bell violation but does not saturate Tsirelson. That the orbit space *does* reach the Tsirelson bound under the ideal SU(2) generator confirms it supports the full quantum correlation range; the limitation is in the graph-derived rotation, not in the orbit Hilbert space.

7.2 Topology determines violation

An ADE sweep across 12 graph types reveals (P13 T3):

ADE type	Aut type	$\mathcal{NC}$	$S_{\max}$	Bell violation?
A_2	$\mathbb{Z}_2$	0	—	No (1 orbit)
A_3	$\mathbb{Z}_2$	0	2.61	Yes
A_4	$\mathbb{Z}_2$	0	2.69	Yes
D_4	S_3	1.22	2.51	Yes
E_7	trivial	0	1.97	No
E_8	trivial	0	1.98	No

Bell violation requires nontrivial automorphisms ($|\text{Aut}| > 1$): graphs with trivial automorphisms (E_7, E_8) do not violate Bell inequalities. The topology of the graph determines whether quantum correlations exceed classical bounds. This is a necessary condition, not a sufficient one: several nontrivial-Aut types in the full sweep (A_5 – A_7, D_5, D_6, E_6) also fall below $S = 2$; the table above lists representative types.

7.3 The Fibonacci connection

A_4 's orbit Laplacian gives $S_{\max} = 2.685$. This matches the independently-derived Fibonacci Bell parameter $S_{\text{Fibonacci}} = 2.683$ (from PAC Bell analysis, scripts 23–32 in `pac\_confluence\_xi`) to 0.09%. The Fibonacci-weighted entanglement discovered through PAC tree analysis is the same as the entanglement produced by A_4 's orbit structure.

8. The Schrödinger equation from the orbit Laplacian

8.1 The Hamiltonian

The orbit Laplacian is:

$$H = B^T L B$$

where $L = D - A$ is the graph Laplacian (degree minus adjacency) and B is the orbit basis matrix (M14 exp_01 columns are the orbit basis vectors).

This single matrix serves three roles:

1. **Hamiltonian:** generates time evolution $U(t) = e^{-iHt}$
2. **Bell generator:** rotates measurement bases via $R(\theta) = e^{i\theta G}$ where $G \propto H$
3. **Path counter:** H^n counts n -hop paths on the orbit graph

8.2 Properties verified (P14)

Unitarity (T1): $U(t)^\dagger U(t) = I$ to 10^{-15} for all t , all ADE types tested.

Equivariance (T2): $[P_g, H] = 0$ for all $g \in \text{Aut}(G)$ — gauge invariance is structural, not imposed. Verified for all ADE types.

Conservation (T3): Energy $\langle \psi | H | \psi \rangle$ and probability $\sum_k |c_k|^2$ are both conserved under time evolution. Energy drift: $< 2 \times 10^{-15}$.

Energy spectrum (T4): A_5 's orbit eigenvalue ratio $(2 + \varphi)/(3 - \varphi) = \varphi^2$ exactly — from the algebraic identity $\varphi^2 = \varphi + 1$. The golden ratio appears in the energy spectrum.

Time-energy uncertainty (T5): Mandelstam-Tamm bound $\Delta E \cdot \tau_\perp \geq \pi/2$ verified to 0.1% on orbit space.

Rabi oscillations (T6): D_4 shows orbit transitions with period π , maximum transfer 75% = 3/4 (from 3 leaves out of 4 vertices), detuning 2, coupling $\sqrt{3}$.

9. The path integral on orbit space

9.1 Propagator decomposition (P15)

The propagator $K(t) = e^{-iHt}$ can be expanded:

$$K(t) = \sum_{n=0}^{\infty} \frac{(-iHt)^n}{n!}$$

Each term H^n counts n -hop paths on the orbit graph. The propagator is a sum over paths, each weighted by $(-it)^n/n!$: this is the Feynman path integral, derived from graph combinatorics.

Short-time verification: $|K(dt) - (I - iH \cdot dt)| < 10^{-5}$ for $dt = 0.001$ across all ADE types tested. Second-order: $|K(dt) - (I - iHdt - H^2 dt^2/2)| < 10^{-8}$. Convergence ratio: $\sim 1000\times$ per order.

9.2 Multi-path interference

On A_5 (3 orbits), the transition $O_1 \rightarrow O_3$ has zero direct coupling ($H_{13} = 0$) but nonzero transition probability — the amplitude goes through O_2 as a virtual intermediate state (P15 T2).

The transition shows constructive/destructive interference with visibility 0.87. This is the graph-theoretic version of virtual particle exchange: O_2 mediates the $O_1 \leftrightarrow O_3$ interaction without ever being "occupied" in the classical sense.

10. Quantum Zeno and anti-Zeno effects

10.1 Zeno freezing (P15 T3)

Frequent measurement of orbit occupation freezes transitions. On D_4 with total time $T = 1.0$:

Measurements N	P_{survive}	Zeno ratio
1 (free)	0.380	1.00
2	0.220	0.58
10	0.740	1.95
100	0.970	2.55
1000	0.997	2.62

At $N = 1000$, the system is frozen with 99.7% probability. Classical systems do not show Zeno freezing.

10.2 Anti-Zeno dip (P15 T4)

At $N = 2$, P_{survive} drops below free evolution (0.220 vs 0.380). This anti-Zeno effect means measurement at the "wrong" rate accelerates decay by 26%. The crossover rate $N^* \sim T \cdot \Delta$ (time $\times$ spectral gap) is topology-dependent — different graphs have different anti-Zeno crossovers.

11. Decoherence and einselection

11.1 The model (P16)

System: D_4 orbit space (2D qubit). Environment: ADE graph orbit space (d_e dimensions). Coupling: spin-boson dephasing $H_{\text{int}} = \lambda \sigma_z \otimes H_{\text{env}}$.

When the system is in $|O_1\rangle$, the environment evolves under $(1 + \lambda)H_{\text{env}}$. When in $|O_2\rangle$, under $(1 - \lambda)H_{\text{env}}$. The environment learns which orbit — decoherence follows.

11.2 Exact decoherence function (P16 T2)

$$|\rho_{12}(t)| = \frac{1}{2} |\langle 0_{\text{env}} | e^{-2i\lambda H_{\text{env}} t} | 0_{\text{env}} \rangle|$$

Numerical match: 1.7×10^{-15} (machine precision). This is not approximate — it is an exact consequence of pure dephasing dynamics on orbit space.

11.3 Fermi golden rule (P16 T3)

At short times: $\text{purity}(t) \approx 1 - 2\lambda^2 \cdot \text{Var}(H_{\text{env}}) \cdot t^2$

The decoherence rate scales as λ^2 (tested across 4 decades, spread 0.09%). The rate is determined by the spectral variance of the environment Hamiltonian in the initial state — a graph-theoretic quantity.

Across 6 ADE environment topologies: $\Gamma^2 / (\lambda^2 \cdot \text{Var}) = 0.999 \pm 0.04\%$. The formula is universal.

11.4 Einselection (P16 T4)

The orbit basis is the pointer basis. Under pure dephasing:

- Orbit eigenstate $|O_1\rangle$: purity = **1.000000** at all times
- Superposition $|+\rangle$: purity drops to **0.514**

The environment selects the orbit basis as preferred. Orbit eigenstates are immune to decoherence; superpositions decay. This is einselection (the environment-induced selection of a preferred basis), derived from graph coupling rather than postulated.

12. Honest limitations

12.1 Scores

- M14: 40/44 (91%)
- P13: 6/6 (100%)
- P14: 6/6 (100%)
- P15: 5/5 (100%)
- P16: 5/5 (100%)
- **Total: 62/66 (94%)**

12.2 The 4 failures (M14)

1. **Graph double-slit** (exp_06: 1/4): Orbit-level interference is algebraic, not positional. Superpositions of disjoint orbits produce vertex-level probabilities that are classical mixtures (no cross-terms). The double-slit requires same-orbit overlap, which the orbit framework doesn't naturally provide. This is a genuine limitation, not a test design issue.
1. **Born rule T4** (exp_03: 3/4): Fibonacci-weighted Born probabilities deviate from orbit-measure predictions for non-uniform initial states. The orbit measure is exact for the uniform state but requires correction for PAC-weighted states.
1. **Orbit interference visibility** (exp_06 T2-T4): The interference mechanism works at the orbit level (algebraic phases) but does not reproduce vertex-level fringe patterns. This is because orbits have disjoint vertex support — there is no spatial overlap for cross-terms.

These failures point to a clear boundary: the orbit framework captures algebraic quantum mechanics (Hilbert space, Born rule, uncertainty, entanglement, Bell violation) but does not naturally produce positional quantum mechanics (double-slit fringes, spatial wavefunctions). The bridge would require a continuum limit of ADE chains, which remains open.

12.3 Structural vs derived

Approximately 30% of M14 results are structural (orbit orthogonality, Gram = identity). The remaining 70% — Born rule verification, D_4 uniqueness, Bell violation, Schrödinger dynamics, Zeno effect, decoherence — are derived consequences that were not guaranteed by the construction.

13. Predictions

13.1 From M14 (12 predictions)

#	Type	Statement
P1	Precise	Quantum uncertainty requires non-abelian $\text{Aut}(G)$: only D_4 among ADE $\leq \text{rank } 8$
P2	Precise	Orbit Gram matrix = identity for ALL ADE types
P3	Precise	Born probabilities = $ O_i /n$ for uniform state
P4	Precise	Trivial irrep multiplicity = number of orbits (Burnside)
P5	Directional	SEC complexification enables full interference range
P6	Directional	Orbit interference is algebraic, not positional
P7	Precise	D_4 triality (S_3) is unique source of non-commutativity
P8	Directional	Minimum uncertainty product nonzero only for D_4
P9	Precise	M13 complement orbits = M14 automorphism orbits
P10	Constraint	Gauge-invariant entanglement requires $ \text{Aut} > 1$
P11	Constraint	Orbit dimension grows monotonically with rank per ADE family
P12	Directional	Real orbit projectors structurally lose phase (SEC arrow)

13.2 From P13–P16 (4 new predictions)

#	Type	Statement
P13	Precise	CHSH > 2 requires nontrivial $\text{Aut}(G)$ on product graphs
P14	Precise	Orbit Laplacian Hamiltonian: $[P_g, H] = 0$ (gauge invariance structural)
P15	Precise	Decoherence rate $\Gamma^2 = \lambda^2 \cdot \text{Var}(H_{\text{env}})$ universal across ADE
P16	Precise	Orbit basis = pointer basis under dephasing coupling

14. Relationship to standard quantum mechanics

QM Postulate	Graph Origin	Paper Section
Hilbert space	Orbit space $L^2(V/\text{Aut}(G))$	§2
Born rule	Orbit volume counting $ O_i /n$	§3
Measurement collapse	Gauge fixing (orbit projection)	§4
Uncertainty principle	Non-abelian Aut (D_4 triality)	§5
Tensor product	Cartesian product graph	§6
Bell violation	Topology + SEC rotation	§7
Schrödinger equation	$i\partial_t \psi\rangle = H \psi\rangle$, $H = B^T L B$	§8
Path integral	Sum over n -hop graph paths	§9
Zeno effect	Projection dynamics on orbit space	§10
Decoherence	Orbit-environment entanglement	§11

Every row is derived, not assumed.

15. Conclusion

Quantum mechanics here is not a separate axiom system; it is what ADE graph symmetries produce when promoted to dynamics.

The orbit Hilbert space resolves the PSD degeneracy of M13, provides a natural Born rule, gives D_4 triality a unique physical role (the source of quantum uncertainty), and produces the full operational toolkit of quantum mechanics: Bell violation, Schrödinger evolution, path integrals, Zeno effect, decoherence, and einselection.

The single remaining gap — positional quantum mechanics (spatial wavefunctions, double-slit fringes) — points toward the continuum limit of ADE chains as the natural next step (M15: Dynamics as Orbit Flow, fully delivered through P13–P16 for the algebraic sector).

Source code: `foundational/experiments/milestone14/` (11 experiments + P13–P16 scripts)

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